Synopsis

Sentiment Analysis of Product Reviews using Machine Learning

Problem Id: 16

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- **Problem Description:**

Sentiment analysis is the process of identifying the emotional tone behind a piece of text. In this project, we aim to develop a machine learning model that classifies product reviews into positive, negative, or neutral categories. With the rapid growth of e-commerce platforms, customers rely heavily on reviews before purchasing products. However, manually analyzing thousands of reviews is time-consuming and inefficient.

This project automates the process by using natural language processing (NLP) and machine learning techniques to extract meaningful insights from textual data. The system preprocesses the text, converts it into numerical features, and applies classification algorithms to predict sentiment. This helps businesses understand customer feedback, improve product quality, and enhance customer satisfaction. The project has real-life applications in online shopping platforms, social media monitoring, and customer service systems.

• Steps of Implementation:

1.Dataset Selection

A dataset containing product reviews and corresponding sentiment labels is selected.

2.Data Preprocessing

• Removal of noise (punctuation, special characters, numbers) using Regular Expressions

• Conversion of text to lowercase to ensure consistency

• Tokenization (splitting sentences into words)

• Removal of stopwords to eliminate less meaningful words

• Lemmatization using WordNetLemmatizer to convert words into their meaningful base form (e.g., ‘cars’ → ‘car’)

3.Feature Extraction

Text data is converted into numerical form using TF-IDF vectorization.

4.Feature Selection

Important features are selected using Chi-Square method with SelectKBest.

5.Train-Test Split

Dataset is divided into training and testing sets (80% training, 20% testing).

6.Model Training

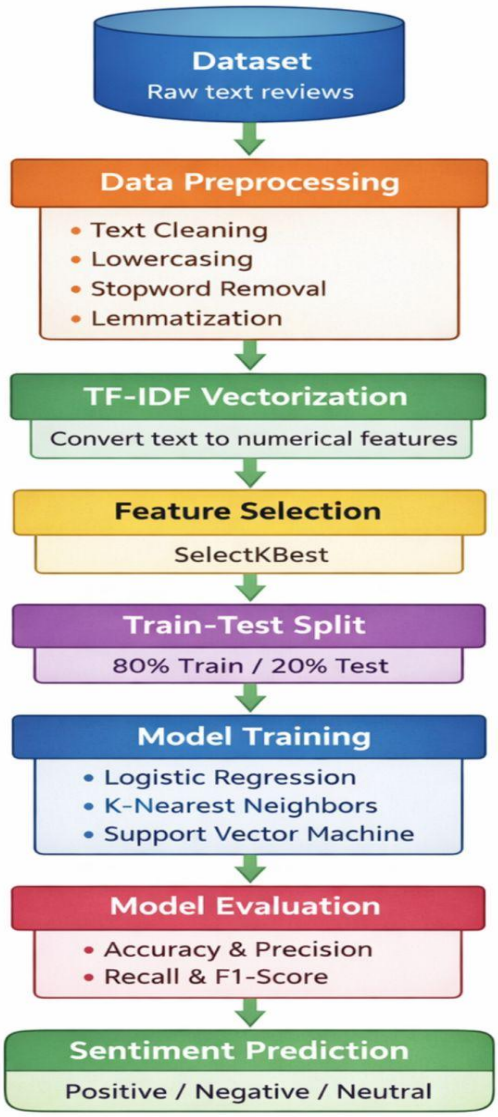
•Logistic Regression

•K-Nearest Neighbors (KNN)

•Support Vector Machine (SVM)

7.Model Evaluation

Models are evaluated using various performance metrics



- **Expected Modules/ Libraries to be used :**

  • **Pandas:** Used for data manipulation and analysis.

  • **NumPy**: Provides support for numerical operations.

  • **NLTK**: Used for text preprocessing (stopwords, lemmatization).

  • **Scikit-learn**: Provides machine learning algorithms and tools.

  • **Re (Regular Expressions):** Used for text cleaning.

  • **Matplotlib**: Used for visualization of results.

- ## Expected Evaluation metrics and Plots to be used:

  • **Accuracy**: Measures overall correctness of the model.

  • **Precision**: Measures the correctness of positive predictions.

  • **Recall:** Measures the ability to detect all positive instances.

  • **F1 Score**: Harmonic mean of precision and recall.

  • **Confusion Matrix:** Shows correct and incorrect predictions.

These metrics help in understanding the efficiency and reliability of the classification models.

- ## Required topics from the subject:

  Strings:
  Used for handling and manipulating textual data such as product reviews.

  Files:
  Used to read and store datasets in CSV format**.**

  Functions:
  Used to modularize code such as text cleaning and preprocessing functions.

  Loops:
  Used for iterating over data during preprocessing and training.

- **Platform Required:**

  Google Colab / Jupyter Notebook / Python IDE

- ## Books and Link Sources:

  1.“Python for Data Analysis” by Wes McKinney

  2. GeeksforGeeks (https://www.geeksforgeeks.org)

  3. Scikit-learn official documentation

  4. Kaggle datasets (https://www.kaggle.com)